

Python input()

1. Question: What is the input() function in Python used for?

Answer: The input() function is used to read input from the user as a string.

2. Question: How can you accept an integer as input from the user using input()?

Answer: You can convert the string returned by input() into an integer using the int() function:

```
num = int(input("Enter an integer: "))
```

3. Question: How do you accept a float input from the user?

Answer: Use float() to convert the string input into a floating-point number:

```
num = float(input("Enter a float: "))
```

4. Question: How can you take multiple space-separated values as input?

Answer: You can use split() to split the input string into a list of values:

```
values = input("Enter values: ").split()
```

5. Question: How do you check if a number entered by the user is positive, negative, or zero?

Answer: Use an if-elif-else block to check the condition:

```
num = int(input("Enter a number: "))
if num > 0:
    print("Positive")
elif num < 0:
    print("Negative")
else:
    print("Zero")
```

6. Question: How do you convert user input to a list of integers?

Answer: After using split(), you can convert each element to an integer using a list comprehension:

```
nums = [int(x) for x in input("Enter numbers: ").split()]
```

7. Question: How do you accept a string input and print it in uppercase?

Answer: You can use the upper() method:

```
user_input = input("Enter a string: ")
print(user_input.upper())
```

8. Question: Write a Python program that accepts a string and prints the number of vowels in it.

Answer:

```
text = input("Enter a string: ")
vowels = "aeiou"
count = sum(1 for char in text if char.lower() in vowels)
```

```
print("Number of vowels:", count)
```

9. Question: Write a program that takes a number as input and checks if it is even or odd.

Answer:

```
num = int(input("Enter a number: "))
if num % 2 == 0:
    print("Even")
else:
    print("Odd")
```

10. Question: How would you check if a string is a palindrome using input()?

Answer:

```
text = input("Enter a string: ")
if text == text[::-1]:
    print("Palindrome")
else:
    print("Not a palindrome")
```

11. Question: Write a program that takes a number as input and prints its square.

Answer:

```
num = int(input("Enter a number: "))
print("Square:", num ** 2)
```

12. Question: Write a program that asks for a number and prints whether it is divisible by 3.

Answer:

```
num = int(input("Enter a number: "))
if num % 3 == 0:
    print("Divisible by 3")
else:
    print("Not divisible by 3")
```

13. Question: How would you check if a number is divisible by both 3 and 7?

Answer:

```
num = int(input("Enter a number: "))
if num % 3 == 0 and num % 7 == 0:
    print("Divisible by both 3 and 7")
else:
    print("Not divisible by both 3 and 7")
```

14. Question: How do you accept a list of comma-separated values as input?

Answer:

```
values = input("Enter comma-separated values: ").split(',')
```

15. Question: Write a Python program that takes two numbers as input and prints their product.

Answer:

```
num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))
print("Product:", num1 * num2)
```

16. Question: Write a program that checks if the input number is a prime number.

Answer:

```
num = int(input("Enter a number: "))
if num > 1:
    for i in range(2, num):
        if num % i == 0:
            print("Not a prime number")
            break
    else:
        print("Prime number")
else:
    print("Not a prime number")
```

17. Question: How can you accept a boolean value (True/False) from the user?

Answer:

```
user_input = input("Enter True or False: ").lower() == "true"
```

18. Question: Write a program that accepts a string and prints the reverse of that string.

Answer:

```
user_input = input("Enter a string: ")
print("Reversed string:", user_input[::-1])
```

19. Question: Write a program that asks for a user's name and age and prints a message.

Answer:

```
name = input("Enter your name: ")
age = int(input("Enter your age: "))
print(f"Hello, {name}. You are {age} years old.")
```

20. Question: Write a program to calculate the factorial of a number using input().

Answer:

```
num = int(input("Enter a number: "))
factorial = 1
for i in range(1, num + 1):
    factorial *= i
print("Factorial:", factorial)
```

21. Question: How do you prevent a user from entering an empty string?

Answer:

```
user_input = input("Enter something: ").strip()
if not user_input:
    print("Input cannot be empty.")
else:
```

```
print(f"You entered: {user_input}")
```

22. Question: Write a program to check if an entered number is a perfect square.

Answer:

```
import math
num = int(input("Enter a number: "))
if math.isqrt(num) ** 2 == num:
    print("Perfect square")
else:
    print("Not a perfect square")
```

23. Question: Write a program that asks the user for a year and determines if it's a leap year.

Answer:

```
year = int(input("Enter a year: "))
if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print("Leap year")
else:
    print("Not a leap year")
```

24. Question: How can you remove leading and trailing spaces from a string input?

Answer:

```
user_input = input("Enter something: ").strip()
```

25. Question: How do you handle incorrect inputs when you expect an integer using input()?

Answer:

```
try:
    num = int(input("Enter a number: "))
except ValueError:
    print("Invalid input! Please enter a valid integer.")
```

26. Question: Write a program that accepts a string and counts the occurrence of a particular character.

Answer:

```
text = input("Enter a string: ")
char = input("Enter a character to count: ")
print(f"Occurrence of {char}: {text.count(char)}")
```

27. Question: How would you convert user input to lowercase using input()?

Answer:

```
user_input = input("Enter a string: ").lower()
```

28. Question: Write a program that accepts a number and prints whether it is a multiple of 10.

Answer:

```
num = int(input("Enter a number: "))
if num % 10 == 0:
    print("Multiple of 10")
else:
    print("Not a multiple of 10")
```

29. Question: How would you check if a string contains only alphabets using input()?

Answer:

```
user_input = input("Enter a string: ")
if user_input.isalpha():
    print("Only alphabets")
else:
    print("Contains non-alphabet characters")
```

30. Question: Write a program to count the number of words in a sentence entered by the user.

Answer:

```
text = input("Enter a sentence: ")
print("Number of words:", len(text.split()))
```

31. Question: How would you accept a date input from the user in Python?

Answer:

```
from datetime import datetime
date_str = input("Enter a date (YYYY-MM-DD): ")
date = datetime.strptime(date_str, "%Y-%m-%d")
print("Entered date:", date)
```

32. Question: Write a program that checks if the entered number is divisible by both 3 and 5.

Answer:

```
num = int(input("Enter a number: "))
if num % 3 == 0 and num % 5 == 0:
    print("Divisible by both 3 and 5")
else:
    print("Not divisible by both 3 and 5")
```

33. Question: Write a program to swap the values of two variables using input().

Answer:

```
a = input("Enter first value: ")
b = input("Enter second value: ")
a, b = b, a
print(f"Swapped values: a = {a}, b = {b}")
```

34. Question: Write a program to take user input and print it without spaces between words.

Answer:

```
user_input = input("Enter a string: ")
print(user_input.replace(" ", ""))
```

35. Question: How do you validate if an entered input is a valid email address?

Answer:

```
import re
email = input("Enter email: ")
if re.match(r"^[^@]+@^[^@]+\.[^@]+$", email):
    print("Valid email")
else:
    print("Invalid email")
```

36. Question: Write a program that accepts a number and prints its cube.

Answer:

```
num = int(input("Enter a number: "))
print("Cube:", num ** 3)
```

37. Question: How would you accept and store multiple names from the user?

Answer:

```
names = input("Enter names separated by commas: ").split(',')
print("Names:", names)
```

38. Question: How would you extract numbers from a string entered by the user?

Answer:

```
import re
text = input("Enter a string: ")
numbers = re.findall(r'\d+', text)
print("Extracted numbers:", numbers)
```

39. Question: How do you find the maximum number from a list of integers entered by the user?

Answer:

```
numbers = list(map(int, input("Enter numbers separated by spaces: ").split()))
print("Maximum number:", max(numbers))
```

40. Question: How would you prompt the user for input until they enter a valid number?

Answer:

```
while True:
    try:
        num = int(input("Enter a valid number: "))
        break
    except ValueError:
        print("Invalid input, please enter a number.")
```

41. Question: Write a program to check if the entered string has digits.

Answer:

```
user_input = input("Enter a string: ")
if any(char.isdigit() for char in user_input):
    print("Contains digits")
```

```
else:  
    print("No digits")
```

42. Question: Write a program to check if the entered string has only whitespace characters.

Answer:

```
user_input = input("Enter a string: ")  
if user_input.isspace():  
    print("Only whitespace")  
else:  
    print("Contains non-whitespace characters")
```

43. Question: Write a program to find the sum of all digits in a string entered by the user.

Answer:

```
text = input("Enter a string: ")  
digit_sum = sum(int(digit) for digit in text if digit.isdigit())  
print("Sum of digits:", digit_sum)
```

44. Question: Write a program that accepts a number and prints its absolute value.

Answer:

```
num = int(input("Enter a number: "))  
print("Absolute value:", abs(num))
```

45. Question: How would you check if a string entered by the user contains any uppercase letters?

Answer:

```
user_input = input("Enter a string: ")  
if any(char.isupper() for char in user_input):  
    print("Contains uppercase letters")  
else:  
    print("No uppercase letters")
```

46. Question: Write a program that converts Celsius to Fahrenheit.

Answer:

```
celsius = float(input("Enter temperature in Celsius: "))  
fahrenheit = (celsius * 9/5) + 32  
print(f"Temperature in Fahrenheit: {fahrenheit}")
```

47. Question: Write a program to find the average of a list of numbers entered by the user.

Answer:

```
numbers = list(map(int, input("Enter numbers separated by space: ").split()))  
print("Average:", sum(numbers) / len(numbers))
```

48. Question: Write a program to count the number of consonants in a string entered by the user.

Answer:

```
text = input("Enter a string: ")
consonants = "bcdfghjklmnpqrstvwxyz"
count = sum(1 for char in text.lower() if char in consonants)
print("Number of consonants:", count)
```

49. Question: How do you check if a string entered by the user contains any punctuation?

Answer:

```
import string
text = input("Enter a string: ")
if any(char in string.punctuation for char in text):
    print("Contains punctuation")
else:
    print("No punctuation")
```

50. Question: Write a program that accepts a sentence and prints the longest word.

Answer:

```
text = input("Enter a sentence: ")
words = text.split()
longest_word = max(words, key=len)
print("Longest word:", longest_word)
```
